RNN applications

§1. Logistic map

- A discrete dynamical system described by the difference eqn.

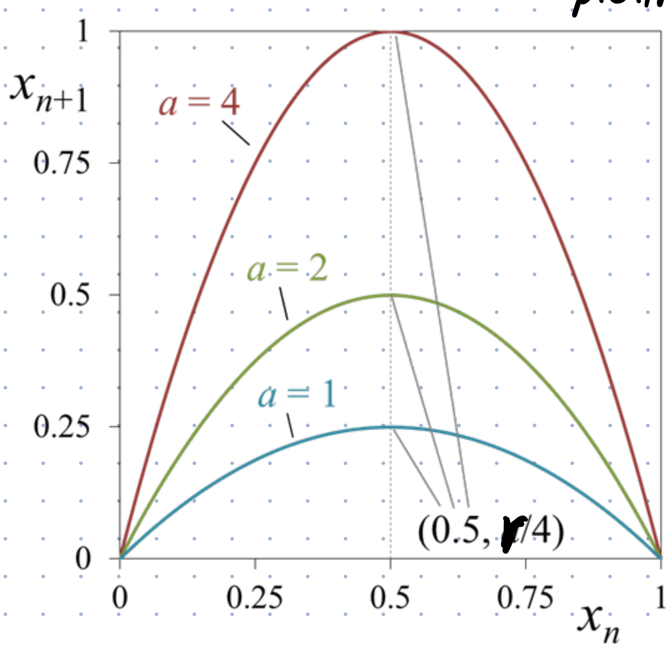
$$x_{n+1} = r\,x_n(1-x_n)$$

- An iconic example of how complex, chaotic behavior can arise from very simple nonlinear dynamical equations

- $x \in [0, 1]$
  $r \in [1, 4]$

- A map $f: x \mapsto rx(1-x)$

- The map can be graphed by plotting the function $f(x) = rx(1-x)$
  The peak is located at $x = \frac{1}{2}$, where $f(\frac{1}{2}) = \frac{r}{4}$
  If $0 \le r \le 4$, then $x \in [0, 1]$.

- 

- The dynamical system generates a sequence (an orbit) starting from an initial value x_0

$$x_1 = f(x_0) = r x_0 (1 - x_0)$$

$$x_2 = f(x_1) = r x_1 (1 - x_1)$$

$$x_3 = f(x_2) = r x_2 (1 - x_2)$$

$$x_4 = \dots$$

- The logistic map may become chaotic.

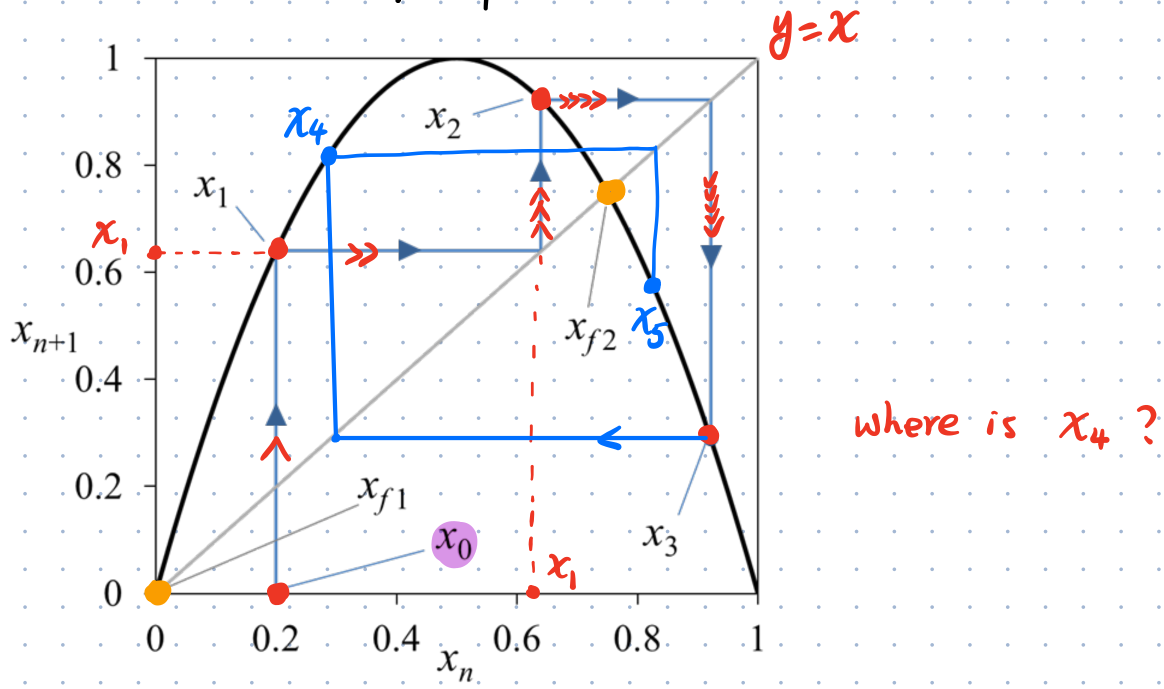
- <u>fixed points</u>:

$$r x (1 - x) = x$$

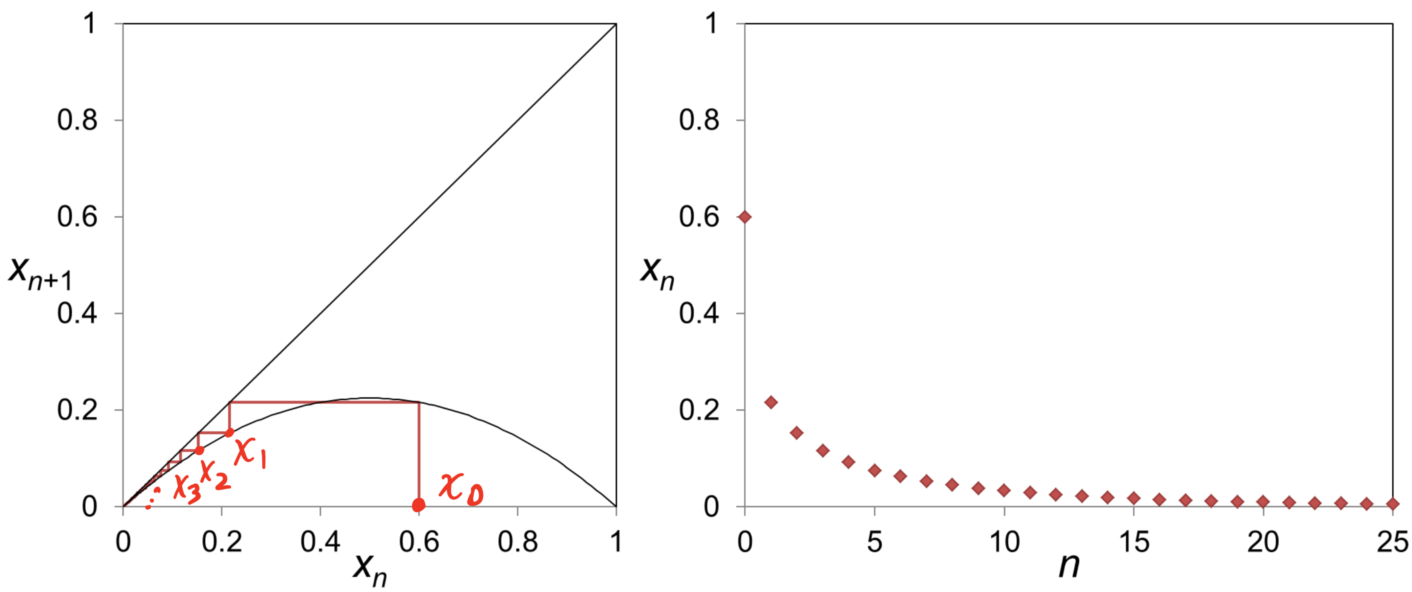
$$\Rightarrow \begin{cases} x = 0 \equiv x_{f_1} \\ \text{or} \quad x = 1 - \frac{1}{r} \equiv x_{f_2} \end{cases}$$

see orange dots below

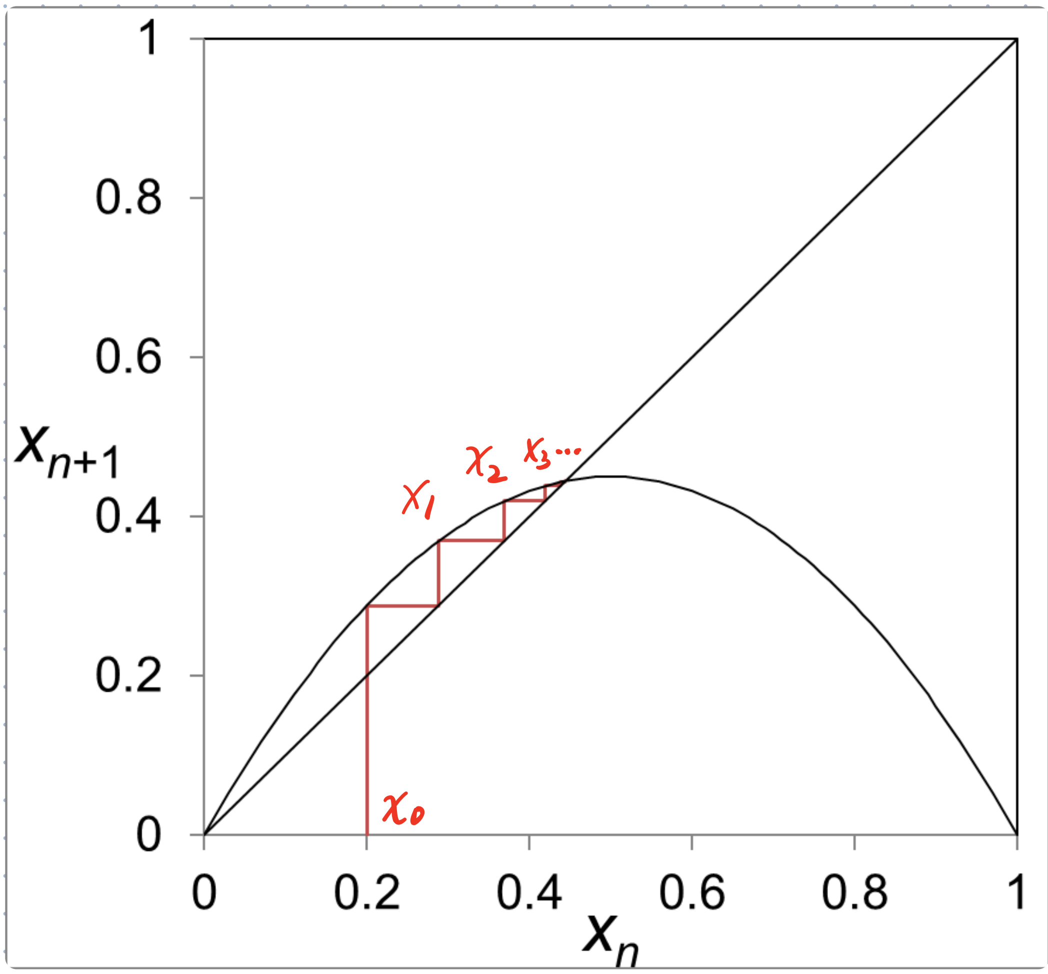
$r = 4$



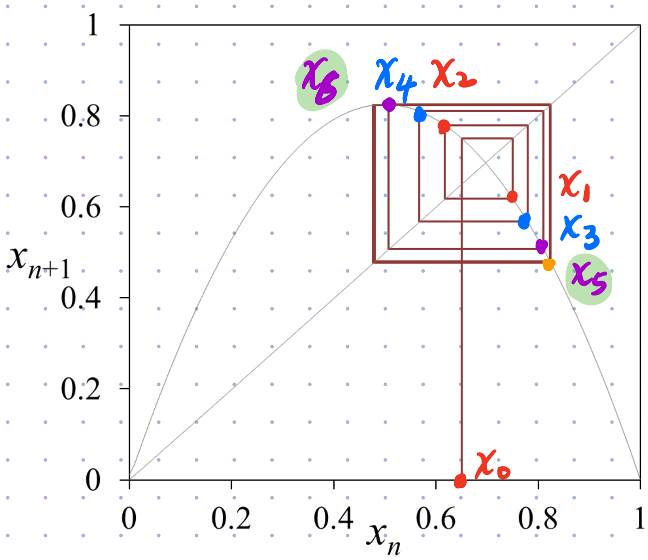
for $r=4$, the mapping doesn't repeat, $\rightarrow$ chaos.

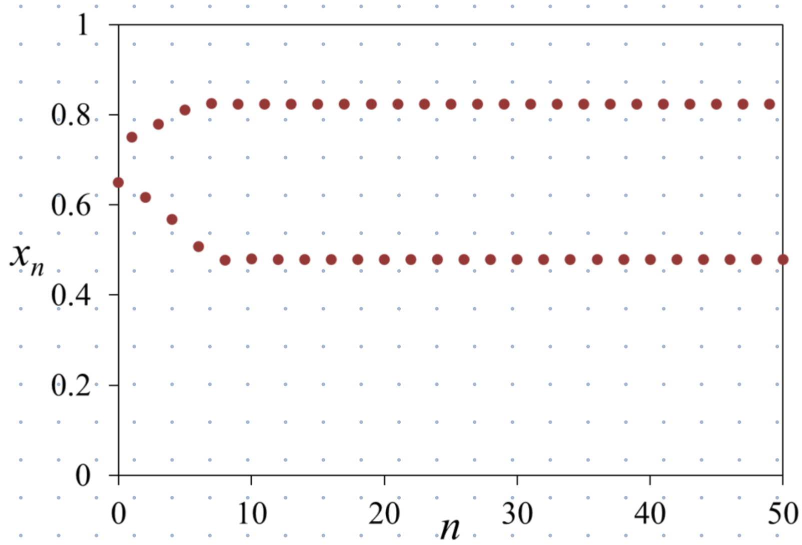


for small r, e.g. $r = 0.9$, the trajectory converges to fixed points, here, $x=0$



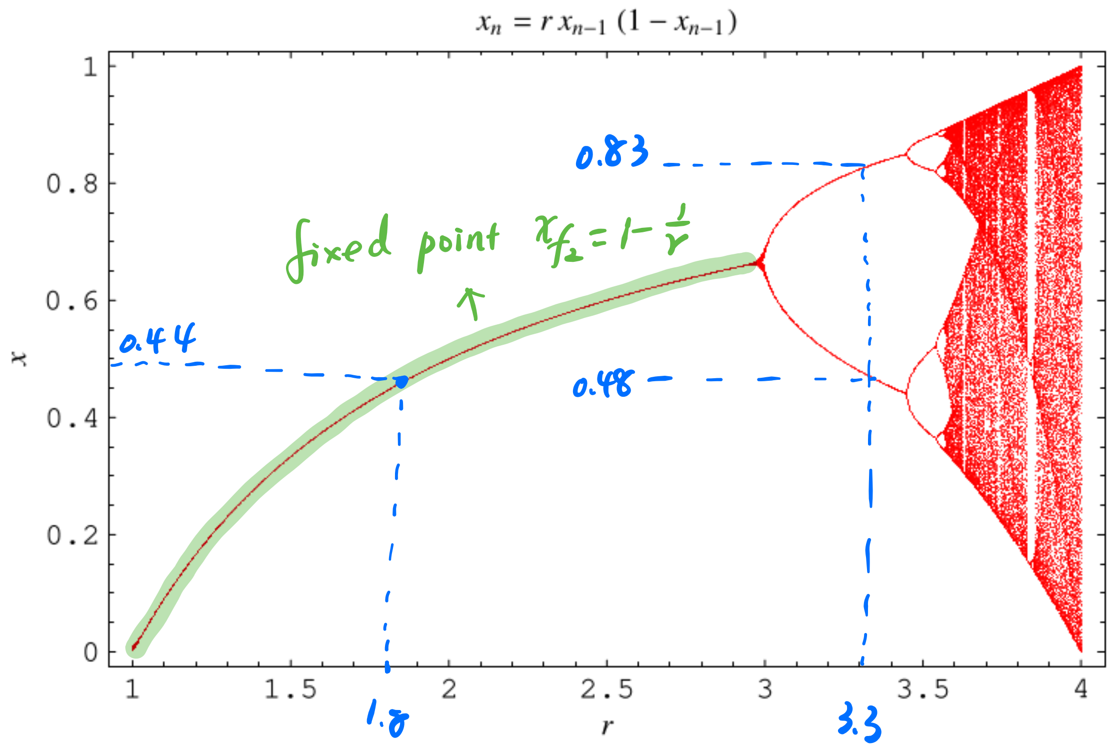
Here, $r = 1.8$. $x_0 = 0.2$, the mapping converges to the other fixed points $x = 1 - \frac{1}{r} = 0.444\cdots$





Here, $r = 3.3$. The orbit cycles through the same values periodically. Here, there are two periodic points $\begin{cases} \approx 0.48 \\ \approx 0.83 \end{cases}$

For larger r, there can be four, eight, ..., peridic points.



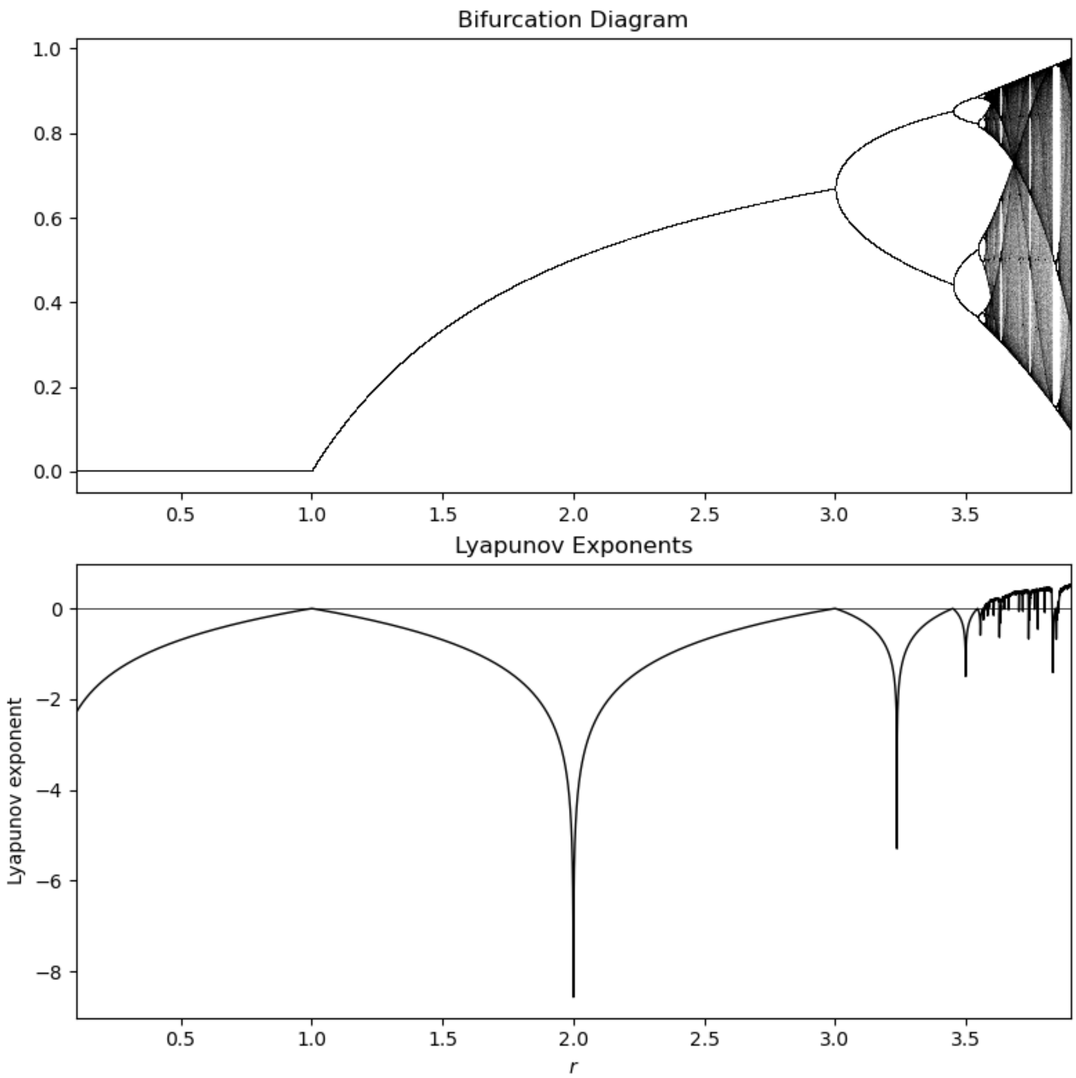
For r close to 4, chaos. $\longrightarrow$ very sensitive to initial condition

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{i=0}^{N-1} \ln |f'(x_i)|$$

It characterizes the distance between two orbits.

$\lambda > 0$: the system is sensitive to initial condition

$\lambda \leq 0$: the system is not sensitive to initial condition



Summary

- Logistic map is deterministic
- Simple equation → complex behavior
- For certain r, system becomes chaotic (chaos is not randomness)
- Tiny change in initial condition → huge divergence.
- Short-term predictable, long-term unpredictable

* Can a neural network approximate chaotic dynamics?
* What does it mean to learn chaos?
* Can we predict the future of chaotic dynamics?
* Logistic map is a discrete-time dynamical system. It evolves step by step. Each step depends on previous value.
* This is exactly the structure RNN is designed for.
  - It models temporal dependence
  - Hidden state acts like internal memory.
  - It can approximate nonlinear mapping.
* develop a RNN to learn the underlying dynamical law of the logistic map.